Volatility, Co-Movement, and Risk in U.S. Retail Stocks: A Time-Series Study of Walmart, Costco, Best Buy, Target, and Kroger (2010–2024)

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Abstract

This paper analyzes daily stock returns for Walmart, Costco, Best Buy, Target, and Kroger from 2010 to 2024 using standard time-series tools. It first tests stationarity and estimates ARMA or ARIMA models to characterize short-run return dynamics, then fits ARCH and GARCH specifications to model time-varying volatility. Cross-stock interactions are examined through Engle–Granger cointegration and a DCC-GARCH framework for dynamic correlations. The results show that Walmart's returns are nearly white noise with moderate volatility persistence, Costco's volatility is driven by short-lived ARCH shocks, and Kroger exhibits the strongest ARMA predictability and event-driven volatility. Best Buy displays high, cyclical volatility tied to discretionary technology demand, while Target's behavior lies between Walmart and Best Buy. Cointegration and crisis-time correlation spikes imply limited diversification within retail during stress, with implications for portfolio construction and risk management.

Keywords: retail stocks, ARMA, ARIMA, ARCH, GARCH, DCC-GARCH, cointegration, volatility clustering

I. Introduction

I.1 Background and Motivation

From 2010 to 2019, the U.S. retail sector transitioned from post-crisis repair to the full force of e-commerce disruption. In the early part of the decade, firms were emerging from the 2008–2009 financial crisis into a period of cautious but steadily improving consumer demand, tighter inventory management, and more conservative balance sheets. As the recovery took hold, competition increasingly shifted away from traditional brick-and-mortar rivalries and toward online platforms. Amazon's growth forced incumbents such as Walmart, Target, Costco, Kroger, and Best Buy to invest heavily in digital channels, curbside pickup, and last-mile delivery. Firms that adapted quickly—Walmart with online grocery, Target with same-day fulfillment, and Best Buy with price matching and Geek Squad—stabilized or grew market share, while laggards exited, restructured, or were acquired.

The COVID-19 pandemic then introduced an entirely different shock. Beginning in March 2020, panic buying and stay-at-home orders created unprecedented surges in demand for groceries, household staples, and home electronics. Walmart, Costco, and Kroger benefited from essential-retail status but faced severe stress in supply chains and labor costs. Best Buy saw a boom in sales of laptops, monitors, and home entertainment as remote work and online schooling took hold, but its stores were temporarily closed or capacity constrained. Target, with a strong suburban footprint and omnichannel capabilities, became one of the clearest winners as households shifted spending from services to goods. Since 2022, the sector has had to deal with a more traditional macroeconomic headwind: high inflation, rapid interest rate increases, and post-pandemic normalization. Inventory gluts, markdowns, and a shift in spending back toward travel and services forced Walmart and Target to issue profit warnings, while grocery chains like Kroger grappled with higher input costs and political scrutiny over alleged "greedflation."

Against this backdrop, this paper focuses on five industry leaders that represent distinct sub-sectors of the retail economy. Walmart Inc. (WMT) is the canonical discount mass-merchandise retailer with a large grocery mix and global footprint. Costco Wholesale Corporation (COST) operates a membership warehouse model built around bulk sales and recurring fee income. The Kroger Co. (KR) is a primarily U.S.-focused supermarket chain with strong regional brands. Best Buy Co., Inc. (BBY) is a specialty electronics retailer whose fortunes are tied to discretionary technology cycles. Target Corporation (TGT) is an upscale discount retailer positioned as "cheap chic," with a mix of groceries, apparel, and home goods aimed at higher-income suburban consumers. All five firms are exposed to broad consumer spending trends and macroeconomic shocks, but their business models, customer bases, and strategic choices imply different risk–return trade-offs.

I.2 Research Objectives

The goal of this paper is to use time-series econometric tools to understand how these five retail stocks behave both individually and as a group. Using daily data from January 2010 to December 2024, the analysis first establishes whether each return series is stationary using Augmented Dickey–Fuller (ADF), Phillips–Perron (PP), and, where appropriate, Kwiatkowski–Phillips–Schmidt–Shin

(KPSS) tests. It then estimates ARMA or ARIMA models of the conditional mean to detect any short-run predictability in returns. Because financial returns typically exhibit volatility clustering, the paper next fits ARCH and GARCH models to the residuals to capture time-varying conditional variance and measure volatility persistence. With the univariate behavior characterized, the paper turns to cross-stock interactions through Engle–Granger cointegration and a DCC-style multivariate GARCH framework for dynamic correlations. Finally, the analysis translates these statistical results into portfolio and risk-management implications for investors who hold or are considering exposure to the U.S. retail sector.

I.3 Significance of the Study

This study has practical value for both portfolio managers and risk managers. Volatility persistence estimates, given by the sum $\alpha + \beta$ in GARCH models, provide a quantitative measure of how long shocks to risk last, which can be used to scale positions dynamically rather than assuming constant volatility. Cointegration results indicate which pairs of retail stocks share a common long-run trend and therefore provide less diversification than their raw correlation might suggest. Conditional volatility forecasts can improve Value-at-Risk calculations relative to simple historical standard deviations and, when combined with dynamic correlations, can support better stress testing during crisis periods.

The analysis also illustrates how econometric models line up with well-known events in the retail sector. The largest spikes in conditional volatility for Walmart and Kroger coincide with the March 2020 COVID-19 crash and the 2022–2024 inflation and merger episodes. For Costco, volatility shocks appear sharp but short-lived, consistent with a highly predictable membership model that reassures investors quickly. For Best Buy, high-volatility regimes extend around its turnaround, pandemic boom, and post-COVID normalization. By explicitly connecting ARMA and GARCH results to these events, the paper bridges the gap between formal time-series diagnostics and the narrative investors use to interpret retail stocks.

I.4 Report Structure

The remainder of the paper proceeds as follows. Section II describes the data, the construction of daily returns, and basic descriptive statistics. Section III outlines the methodology, focusing on stationarity tests, ARMA or ARIMA modeling, ARCH or GARCH volatility modeling, and the logic of cointegration and dynamic correlations. Section IV presents the empirical results stock by stock: Costco, Kroger, Target, Walmart, and Best Buy, followed by cointegration and multivariate GARCH evidence. Section V combines cross-stock patterns, discusses co-movement and crisis-time behavior, and draws out portfolio implications. Section VI concludes with recommendations and future extensions.

II. Data Description

II.1 Data Source and Sample

All price data are obtained from the Bloomberg Terminal. For each company, the sample uses the daily adjusted closing price for the primary U.S. listing: Walmart (WMT US Equity), Costco (COST US Equity), Best Buy (BBY US Equity), Target (TGT US Equity), and Kroger (KR US Equity).[1] The sample runs from January 4, 2010, the first trading day of that year, to December 31, 2024. After excluding days with missing data, each series contains approximately 3,900 to 3,913 trading days.

II.2 Return Construction

For each firm, daily logarithmic returns are computed as

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right), \tag{1}$$

where P_t denotes the adjusted closing price on day t.

Log returns are preferred because they are time-additive, scale-free, and more symmetric than simple percentage changes. This transformation also helps stabilize variance, which is desirable for linear time-series modeling. The return series are tested directly for stationarity, and differencing is used only when the unit-root results fail to reject non-stationarity.

II.3 Descriptive Statistics

Table 1 reports summary statistics for daily returns from 2010 to 2024. Mean daily returns are small but positive for all stocks, ranging from 0.0200% for Best Buy to 0.0700% for Costco. Walmart has the lowest unconditional volatility, with a daily standard deviation of about 1.19%, while Best Buy is the most volatile at roughly 2.42%. All five stocks display negative skewness and very high kurtosis, which is consistent with the standard stylized facts of financial returns (Fama, 1970). Jarque–Bera tests strongly reject normality in every case.

[1] Adjusted close prices incorporate splits, cash dividends, and other corporate actions, so they better reflect the total return to a buy-and-hold investor.

Table 1. Descriptive Statistics for Daily Returns, 2010–2024

Statistic	Walmart	Costco	Best Buy	Target	Kroger
Observations	3,912	3,912	3,913	3,912	3,912
Mean (%)	0.0415	0.0700	0.0200	0.0263	0.0456
Std. Dev. (%)	1.1895	1.2605	2.4200	1.7744	1.6117
Minimum (%)	-12.07	-13.29	-33.67	-28.67	-20.93
Maximum (%)	11.07	9.49	19.46	18.59	10.98
Skewness	-0.2083	-0.4796	-0.7077	-1.4660	-0.8084
Kurtosis	19.72	12.58	21.41	41.41	17.21
Jarque–Bera	45,601.54	15,124.25	55,568.87	241,974.40	33,357.36
JB p-value	0.0000	0.0000	0.0000	0.0000	0.0000

III. Methodology

This section introduces the econometric tools used in the analysis. The focus is on economic interpretation and application to the retail stock data rather than on formal proofs.

III.1 Stationarity Testing

Before fitting ARMA or GARCH models, each stock's daily returns are tested for stationarity. A stationary series has constant mean and variance, and autocovariances that depend only on lag length. Non-stationary series with unit roots can generate spurious regressions and misleading inference. For each stock, the analysis applies an Augmented Dickey–Fuller (ADF) test, a Phillips–Perron (PP) test, and a KPSS test (Said and Dickey, 1984; Phillips and Perron, 1988; Kwiatkowski, Phillips, Schmidt, and Shin, 1992). If the ADF and PP tests reject a unit root and the KPSS test does not reject stationarity, the return series is treated as stationary in levels. Otherwise, the analysis considers first differences and ARIMA specifications.

III.2 ARMA and ARIMA Models for the Mean

To capture short-run predictability in returns, the paper estimates ARMA(p, q) or ARIMA(p, d, q) models. An AR(p) model expresses the current return as a linear combination of its own lags. An MA(q) model expresses the current return as a function of current and lagged shocks. ARMA models combine both dynamics and are estimated by maximum likelihood (Box, Jenkins, Reinsel, and Ljung, 2015). Model identification begins with the autocorrelation function (ACF) and partial autocorrelation function (PACF), and low-order specifications are compared using the Akaike Information Criterion (AIC) and Schwarz or Bayesian Information Criterion (BIC).

III.3 ARCH and GARCH Volatility Models

The most salient empirical property of financial returns is volatility clustering: large absolute returns tend to follow large absolute returns, and quiet periods tend to follow quiet periods. In a

GARCH$(1,1)$ model, the mean equation takes the form

$$r_t = \mu + \text{ARMA terms} + \varepsilon_t, \qquad \varepsilon_t = \sigma_t z_t, \tag{2}$$

where z_t is an iid innovation term. The conditional variance evolves as

$$\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2. \tag{3}$$

Here, $\omega > 0$ is the baseline variance, α captures the immediate effect of recent shocks, and β captures volatility persistence (Engle, 1982; Bollerslev, 1986). The key measure is $\alpha + \beta$: values close to one imply slow decay of volatility shocks, while lower values imply faster mean reversion.

ARCH(q) models are a special case in which variance depends only on lagged squared residuals:

$$\sigma_t^2 = \omega + \alpha_1\varepsilon_{t-1}^2 + \cdots + \alpha_q\varepsilon_{t-q}^2. \tag{4}$$

The paper compares low-order ARCH and GARCH specifications using information criteria and interpretability.

III.4 Cointegration and Dynamic Correlations

Beyond individual behavior, the paper examines how the stocks move together over the long run and during periods of market stress. Cointegration formalizes the idea of a stable long-run relationship among non-stationary price levels. The analysis uses the Engle–Granger two-step procedure, which first estimates a cointegrating regression and then tests the residuals for stationarity (Engle and Granger, 1987).

Dynamic correlations are interpreted through a DCC-GARCH framework. In this setting, conditional correlations evolve over time and often rise during crises, reducing diversification exactly when it is most needed (Engle, 2002). Although the paper does not derive the full estimator, it uses the estimated DCC parameters to discuss persistence in joint risk and co-movement.

III.5 Forecasting Scope and Limitations

Although ARMA and GARCH models are widely used for return and volatility forecasting, this report does not present formal out-of-sample forecasts. Producing reliable forecasts would require systematic model comparison across many alternative specifications, rolling or expanding estimation windows, and rigorous forecast evaluation. Because the models here are intentionally simple and estimated on a single full-sample window, the emphasis remains on in-sample dynamics, volatility behavior, and cross-stock comparisons rather than on forecast performance.

IV. Data Analysis

IV.1 Costco

Costco Wholesale Corporation operates a membership-based warehouse club model, charging annual fees in exchange for access to bulk merchandise at minimal markups. This business model

creates recurring revenue and customer lock-in, which differentiates Costco from traditional retailers.

Table 2 reports Costco's return unit-root tests. The paper ultimately models Costco using an ARIMA$(1, 1, 1)$ specification after comparing low-order alternatives and examining correlograms. In volatility modeling, the evidence points to strong but short-lived ARCH effects. The ARCH(2) model outperforms the GARCH$(1, 1)$ model on log-likelihood and information criteria, which suggests that Costco's volatility is driven more by short-term shocks than by highly persistent conditional variance.

Table 2. Costco Return Unit Root Tests

Test	Specification	Test Statistic	1% Crit.	5% Crit.	10% Crit.	p-value	Conclusion
ADF	Constant & Trend	-63.99	-3.96	-3.41	-3.13	0.0000	Reject unit root
ADF	Level only	-63.79	-2.57	-1.94	-1.62	0.0001	Reject unit root
ADF	First Difference	-24.71	-2.57	-1.94	-1.62	0.0000	Reject unit root

Table 3. Costco ARCH(2) vs. GARCH(1,1)

	ARCH(2)	GARCH(1,1)
Log Likelihood	23050.70	9875.71
AIC	-12.22	-5.23
SC	-12.22	-5.23
HQC	-12.22	-5.23

For Costco, the ARCH(2) model clearly outperforms the GARCH$(1, 1)$ model across all criteria, confirming volatility clustering driven mainly by short-term shocks rather than long-memory variance persistence.

IV.2 Kroger

Kroger's returns are stationary according to the ADF, PP, and KPSS tests in Table 4. The preferred conditional mean model is ARMA$(2, 1)$, although competing ARMA specifications have very similar information criteria. The evidence of ARCH effects is strong, and the GARCH$(1, 1)$ model is preferred for interpretation because it captures both immediate shocks and persistent variance dynamics.

Table 4. Kroger Return Unit Root Tests

Test	Null Hypothesis	Test Statistic	Prob.	1% Level	5% Level	10% Level
ADF	Unit root (non-stationary)	-26.714	0.000	-2.566	-1.941	-1.617
PP	Unit root (non-stationary)	-61.203	0.000	-3.960	-3.411	-3.127
KPSS	Stationary around constant	0.063	—	0.216	0.146	0.119

Table 5. Kroger ARMA Model Comparison

ARMA Model	AR p-value	MA p-value	AIC	SIC	Adj. R^2
ARMA(1,1)	0.0083	0.000	-5.415	-5.410	0.488
ARMA(2,1)	0.0001	0.000	-5.417	-5.412	0.489
ARMA(3,1)	0.0947	0.000	-5.415	-5.410	0.489

Table 6. Kroger ARCH Model

Variable	Coefficient	Std. Error	Prob.
C	0.000213	1.73E-05	0.000
Resid$(-1)^2$	0.171219	0.016000	0.000
Resid$(-2)^2$	0.007535	0.016000	0.638
Adj. R^2		0.029321	
Log likelihood		21351.8	
AIC		-10.922	
SIC		-10.918	

Table 7. Kroger GARCH(1,1) Variance Equation

Variable	Coefficient	Std. Error	Prob.
C	0.000330	0.000114	0.0038
Resid$(-1)^2$	0.150000	0.050730	0.0031
GARCH(-1)	0.599535	0.130764	0.0000
Adj. R^2		0.401671	
Log likelihood		9460.055	
AIC		-4.8375	
SIC		-4.8295	

Table 8. Kroger ARCH(1) vs. GARCH(1,1)

	ARCH(1)	GARCH(1,1)
Log Likelihood	21351.8	9460.055
Akaike Info Criterion	-10.922	-4.8375
Schwarz Criterion	-10.918	-4.8295
Hannan–Quinn Criterion	-10.921	-4.8347

The sum $\alpha + \beta = 0.7495$ is below one, which means the GARCH model is stable and volatility is

mean-reverting. Kroger therefore combines meaningful short-run predictability in the mean with economically important volatility clustering.

IV.3 Target

Target's return series is stationary according to all three unit-root tests in Table 9. The preferred conditional mean model is ARMA$(1,1)$. The ARCH test points to a modest but statistically significant amount of volatility clustering, and the GARCH$(1,1)$ model captures this time-varying variance more realistically than a pure ARCH model.

Table 9. Target Return Unit Root Tests

Test	Null Hypothesis	Test Statistic	Prob.	1% Level	5% Level	10% Level
ADF	Unit root (non-stationary)	-24.409	0.000	-2.566	-1.941	-1.617
PP	Unit root (non-stationary)	-63.558	0.000	-3.960	-3.411	-3.127
KPSS	Stationary around constant	0.064	—	0.216	0.146	0.119

Table 10. Target ARMA Model Comparison

ARMA Model	AR p-value	MA p-value	AIC	SIC	Adj. R^2
ARMA(1,1)	0.1143	0.000	-5.222	-5.217	0.507
ARMA(2,1)	0.3553	0.000	-5.222	-5.217	0.508
ARMA(3,1)	0.6208	0.000	-5.222	-5.217	0.508

Table 11. Target ARCH(1) Model

Variable	Coefficient	Std. Error	Prob.
C	0.000291	3.31E-05	0.000
Resid$(-1)^2$	0.038126	0.016000	0.0172
Adj. R^2		0.001540	
Log likelihood		18737.75	
AIC		-9.587	
SIC		-9.581	

Table 12. Target GARCH(1,1) Variance Equation

Variable	Coefficient	Std. Error	Prob.
C	0.000416	0.000101	0.0000
Resid$(-1)^2$	0.150000	0.040095	0.0002
GARCH(-1)	0.599535	0.130764	0.0000
Adj. R^2		0.2748	
Log likelihood		8928.849	
AIC		-4.5646	
SIC		-4.5566	

Table 13. Target ARCH(1) vs. GARCH(1,1)

	ARCH(1)	GARCH(1,1)
Log Likelihood	18737.75	8928.849
Akaike Info Criterion	-9.587	-4.5646
Schwarz Criterion	-9.582	-4.5566
Hannan–Quinn Criterion	-9.586	-4.5617

The sum of the ARCH and GARCH terms is approximately 0.75, which implies stable and mean-reverting volatility. Target therefore lies between defensive names such as Walmart and more cyclical names such as Best Buy.

IV.4 Walmart

Walmart is the largest brick-and-mortar retailer in the United States and serves as the defensive anchor of the sample. Unit-root tests strongly support stationarity in levels, as shown in Table 14. The preferred mean model is ARMA$(1,1)$ in levels. In the simple ARMA specification, the mean process is close to white noise, but once volatility is modeled explicitly, the ARMA terms become significant, which suggests hidden mean reversion masked by volatility noise.

Table 14. Stationarity Tests for Walmart Daily Returns

Test	Null Hypothesis	Test Statistic	1% Crit.	5% Crit.	10% Crit.	Conclusion
ADF	Unit root (non-stationary)	-65.693	-3.432	-2.862	-2.567	Reject unit root
PP	Unit root (non-stationary)	-65.896	-3.432	-2.862	-2.567	Reject unit root
KPSS	Stationary around constant	0.195	0.739	0.463	0.347	Do not reject stationarity

Table 15. Walmart ARMA Model Selection

Model	Dependent Variable	AIC	BIC	Log-likelihood	Adj. R^2
AR(1)	WALMART_RETURN	_	_	_	_
MA(1)	WALMART_RETURN	_	_	_	_
ARMA(1,1)	WALMART_RETURN	-6.0251	-6.0240	11,788.31	0.0006
ARMA(1,1)	D(WALMART_RETURN)	-6.0245	-6.0193	11,788.13	0.5242
ARMA(2,1)	D(WALMART_RETURN)	-6.0248	-6.0194	11,788.54	0.5234

Table 16. Walmart GARCH(1,1) Model

Equation	Parameter	Coefficient	z-Statistic	Prob.	Interpretation
Mean	AR(1)	-0.3447	-5.196	0.0000	Significant negative serial correlation
	MA(1)	-0.3597	-5.125	0.0000	Correction for overreaction
Variance	ω (Constant)	0.000193	3.169	0.0015	Baseline long-term variance
	α (ARCH)	0.150000	3.809	0.0001	Sensitivity to recent news
	β (GARCH)	0.599838	5.043	0.0000	Persistence of volatility

For Walmart, the combined persistence measure is high and conditional volatility decays slowly after shocks. This is consistent with Walmart's reputation as a relatively low-beta defensive retailer that still reacts sharply when major macro or firm-specific news arrives.

IV.5 Best Buy

Best Buy is the clearest high-volatility stock in the sample. Its daily returns have near-zero mean but the highest dispersion, strong negative skewness, and very high kurtosis. Unit-root tests indicate that returns are already stationary in levels. The selected mean model is ARMA$(1,1)$, in which both coefficients are statistically significant. The ARCH-LM test strongly rejects homoskedasticity, and the GARCH$(1,1)$ model shows that volatility is persistent but still mean-reverting.

Table 17. Best Buy Descriptive Statistics

Statistic	Value
Observations	3,913
Mean	0.0002
Standard deviation	0.0242
Minimum	-0.3367
Maximum	0.1946
Skewness	-0.7077
Kurtosis	21.4071
Jarque–Bera	55,568.87
JB p-value	0.0000

Table 18. Best Buy Unit Root and Stationarity Tests

Test	Statistic	1% Crit.	5% Crit.	10% Crit.	p-value
ADF	-62.5934	-3.9604	-3.4110	-3.1273	0.0000
PP	-62.5959	-3.9604	-3.4110	-3.1273	0.0000
KPSS	0.0664	0.2160	0.1460	0.1190	—

Table 19. Best Buy ARMA(1,1) Mean Equation

Parameter	Coefficient	Std. Error	t-Statistic	p-value
AR(1)	0.7674	0.3407	2.2523	0.0244
MA(1)	-0.7770	0.3349	-2.3203	0.0204
SIGMASQ	0.000584	0.0000042	139.0776	0.0000

Table 20. Best Buy ARCH(1) LM Test on ARMA(1,1) Residuals

Statistic	Value	p-value
F-statistic	23.6099	0.0000
Obs $\times R^2$	23.4802	0.0000

Variable	Coefficient	Std. Error	t-Statistic	p-value
C	0.000539	0.0000430	12.5142	0.0000
$\text{Resid}^2(-1)$	0.0775	0.0159	4.8590	0.0000

Table 21. Best Buy GARCH(1,1) Model

Parameter	Coefficient	Std. Error	z-Statistic	p-value
AR(1)	-0.0001	6,916.246	-1.9E-08	1.0000
MA(1)	0.0001	6,916.245	1.9E-08	1.0000
$C(\omega)$	0.000380	0.0000621	6.1125	0.0000
Resid$(-1)^2$ (α_1)	0.1500	0.0306	4.9082	0.0000
GARCH(-1) (β_1)	0.6000	0.0628	9.5539	0.0000

Table 22. Best Buy ARCH(1) vs. GARCH(1,1)

	ARCH(1)	GARCH(1,1)
Log Likelihood	17692.34	8705.96
AIC	-9.04	-4.45
SC	-9.04	-4.44
HQC	-9.04	-4.45

Because the goal is to understand time-varying risk rather than merely optimize in-sample fit, the GARCH$(1,1)$ model is retained as the main volatility specification for Best Buy, even though the ARCH(1) model performs slightly better on information criteria.

IV.6 Cointegration

Following the univariate ARCH and GARCH modeling, the analysis conducts cointegration tests on each pair of firms using the Engle–Granger two-step procedure (Engle and Granger, 1987). Table 23 reports the cointegrating regressions, and Table 24 reports the ADF tests on the residuals. Every pair rejects a unit root in the residual spread, which suggests a stable long-run relationship across all five names.

Table 23. Cointegrating Regression Results

Pairs	Coefficient	Std. Error	t-Statistic	p-value	R^2
Costco – Target	0.321	0.009	33.978	0.000	0.2035
Costco – Best Buy	0.160	0.007	22.023	0.000	0.0923
Costco – Walmart	0.572	0.013	42.297	0.000	0.2917
Costco – Kroger	0.293	0.011	26.665	0.000	0.1408
Best Buy – Target	0.519	0.019	27.025	0.000	0.1437
Best Buy – Walmart	0.511	0.029	17.103	0.000	0.0586
Best Buy – Kroger	0.315	0.022	14.079	0.000	0.0429
Target – Walmart	0.637	0.021	30.686	0.000	0.1767
Target – Kroger	0.331	0.016	20.423	0.000	0.0883
Walmart – Kroger	0.275	0.010	27.141	0.000	0.1400

Table 24. ADF Test on Cointegration Residuals

Pairs	ADF Test Statistic	p-value	Stationary	Cointegration
Costco – Target	-64.259	0.000	Yes	Yes
Costco – Best Buy	-64.976	0.000	Yes	Yes
Costco – Walmart	-64.155	0.000	Yes	Yes
Costco – Kroger	-65.615	0.000	Yes	Yes
Best Buy – Target	-62.545	0.000	Yes	Yes
Best Buy – Walmart	-62.275	0.000	Yes	Yes
Best Buy – Kroger	-62.604	0.000	Yes	Yes
Target – Walmart	-62.152	0.000	Yes	Yes
Target – Kroger	-63.319	0.000	Yes	Yes
Walmart – Kroger	-66.447	0.000	Yes	Yes

Among the individual pairs, Target–Walmart appears especially strong, with a slope coefficient of 0.637 and an R^2 of 0.1767. Costco–Walmart is also relatively strong, while Best Buy's relationships with Kroger and Walmart are weaker in explanatory power even though they remain statistically significant.

IV.7 Multivariate GARCH

The paper next summarizes univariate GARCH estimates for each firm and then reports the multivariate DCC-GARCH parameters. These results show substantial and statistically significant long-run correlations across the sample. The dynamic correlation parameters also follow the expected DCC structure, with a small α and a much larger β, which implies slowly evolving and persistent correlations.

Table 25. Univariate GARCH Model Summary by Stock

Stock	AR(1)	MA(1)	Constant	ARCH (α)	GARCH (β)	Log Likelihood
Best Buy	0.60345	-0.620717	0.817362	0.055737	0.805423	9096.710
Costco	-0.83127	0.84251	0.11344	0.08645	0.84195	11799.671
Kroger	-0.813996	0.849320	0.539375	0.181329	0.629710	10769.897
Walmart	0.491808	-0.539960	0.122843	0.087190	0.828150	12022.204
Target	-0.942191	0.934315	0.226317	0.268384	0.737697	10469.763

Table 26. Multivariate DCC-GARCH Model

Parameter	Coefficient	Std. Error	t-Value	p-Value
ρ_{21}	0.260166	0.026339	9.878	0.0000
ρ_{31}	0.196087	0.023030	8.515	0.0000
ρ_{41}	0.205429	0.026470	7.761	0.0000
ρ_{51}	0.351490	0.033171	10.60	0.0000
ρ_{32}	0.360489	0.023530	15.32	0.0000
ρ_{42}	0.480531	0.026566	18.09	0.0000
ρ_{52}	0.400418	0.028066	14.27	0.0000
ρ_{43}	0.353242	0.023211	15.22	0.0000
ρ_{53}	0.287098	0.023101	12.43	0.0000
ρ_{54}	0.392575	0.030486	12.88	0.0000
α	0.032877	0.0083730	3.927	0.0001
β	0.735132	0.084742	8.675	0.0000
Log Likelihood		55910.739		

Because $\alpha + \beta \approx 0.77$, the correlation dynamics satisfy the DCC stationarity condition. The estimates therefore imply moderate but highly persistent positive correlations across the five stocks, which reduces diversification benefits within the retail sector.

V. Discussion

The empirical findings carry several practical implications for investors who hold or are considering positions in these retail stocks.

First, the univariate results highlight clear differences in risk profiles. Walmart emerges as the defensive anchor: it has the lowest unconditional volatility, and its returns are almost white noise, with only mild mean reversion once volatility is controlled for. Costco offers slightly higher volatility but unusually rapid volatility mean reversion, which makes it attractive for investors who care about the duration of high-risk regimes. Kroger has higher volatility and unusually strong ARMA predictability; its stock is influenced heavily by event-driven trading and corporate developments,

especially in the merger period. Best Buy is the clear high-beta, high-volatility name whose risk remains elevated for extended periods after shocks. Target likely sits between Walmart and Best Buy, sharing Walmart's general-merchandise format but with more exposure to discretionary demand and reputational shocks.

Second, GARCH-based volatility forecasts can improve risk management. Rather than assigning each stock a constant volatility number, investors can use conditional variance forecasts to update exposure daily. When Walmart's or Costco's conditional volatility spikes, a risk-averse investor may temporarily scale down position sizes or widen stop-loss thresholds to avoid being stopped out by normal noise in a high-volatility regime. For Kroger and Best Buy, where $\alpha + \beta$ is around 0.75 in several specifications, an investor can anticipate that high volatility will likely last for several weeks after a major shock and adjust risk limits accordingly.

Third, cross-stock behavior matters for portfolio construction. If cointegration tests confirm a long-run relationship between Walmart and Target, owning both may not deliver as much diversification as owning Walmart plus a structurally different name like Costco. If crisis-time correlations spike toward one during systemic shocks, retail-sector diversification will not protect much against market crashes; investors would need to look outside the sector for genuine hedges.

Finally, the strong ARMA structure in Kroger and the spurious ARMA structure in Best Buy under simpler models suggest different roles in active strategies. Kroger's ARMA$(2, 1)$ with an MA coefficient near one points toward potential short-horizon mean-reversion trades around news events. Best Buy's unstable mean dynamics under alternative models argue against naive linear trading rules and instead support direct modeling of volatility and event risk.

VI. Conclusion and Future Work

This paper applied familiar time-series tools—unit-root tests, ARMA or ARIMA models, and ARCH or GARCH volatility models—to daily stock returns for Walmart, Costco, Best Buy, Target, and Kroger over 2010 to 2024. The results show that return series are stationary, volatility clustering is widespread, and co-movement within the retail sector is economically meaningful and persistent.

Three main conclusions emerge. First, retail stocks are not uniform in risk. Walmart and Costco behave as the defensive holdings in the sample, with lower volatility and either moderate or rapid volatility mean reversion. Kroger and Best Buy carry significantly higher risk, with Kroger showing unusually strong ARMA dynamics and Best Buy exhibiting the most extreme tail behavior. Target lies between the defensive and cyclical extremes. Second, ARCH and GARCH models provide a much more realistic description of risk than constant-variance models. Third, long-run comovement and short-run dynamic correlations limit the diversification benefits available within the retail sector, especially during crises when correlations spike.

There are several natural extensions. Asymmetric volatility models such as EGARCH or GJR-GARCH could test whether negative returns increase volatility more than positive returns, which is plausible given the negative skewness of the series. A fully estimated DCC-GARCH model

over rolling windows would quantify time variation in crisis correlations more explicitly. Higher-frequency data could also help isolate firm-specific earnings shocks from broader macro shocks.

Even without these extensions, the main message is clear: retail is not a single, homogeneous risk factor. Within this sector, investors can choose between defensive names with lower and faster-reverting volatility and more aggressive names whose risk and short-run dynamics are strongly event-driven.

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